Autodesk®
Scaleform®

Scaleform 3D

Scaleform 3.2 3D

Mustafa Thamer
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Autodesk®
GAMEWARE

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Autodesk® Scaleform® 4.3

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Autodesk Scaleform

	Scaleform 3D
	Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
	www.scaleform.com
Email	info@scaleform.com
	+1 (301) 446-3200
Fax	+1 (301) 446-3199

目 次

1 概要

Scaleform® 3Di™ 3D Flash
UI

Scaleform Scaleform 3.2 ActionScript
3D Scaleform 3.2 ActionScript 2.0
AS 3.0 3D AS

_z _zscale _xrotation _yrotation _matrix3d _perspfov
HUD UI 3D
3D

3D Scaleform Direct Access API C++ API
3D GFx::Value

3D 3D Flash

2 Flash 10 との類似性

Scaleform 3Di Flash 10 3D
3Di Scaleform AS2

2.1 制約事項

Flash 10 3D 3Di

1.

2.

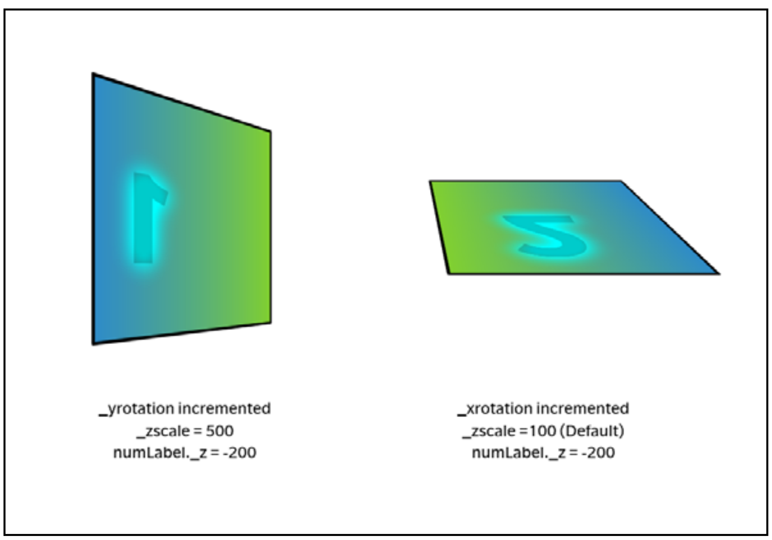

図 1: カリングされずにオブジェクトの後ろ側で描画される裏向き数字の例

2.2 実装

Scaleform 3Di
Flash 10 3D

2D 3D 2D 3D
3D
Flash 10

3 3D での Flash の表示

Scaleform 3.2 Flash 3D 2

3.1 レンダーからテクスチャへの変換

Flash
3D Scaleform 3.2 Flash 3D

3Di Flash
3D 3D
3D Z
UI

Flash
Scaleform
Flash

Scaleform SDK Browser SWF to Texture

3.2 3Di

Scaleform 3.2 3Di ActionScript DirectAccess API
C++ 3Di

Scaleform
Scaleform

3Di Scaleform
UI
UI

UI UI 3D
UI

4 3Di API

4.1 ActionScript 2 API

3Di Flash 3

2 3D
3D x y z
3D 3 3D
3D null
ActionScript foo._matrix3d = null 2D
z 0 3D

2D z 3D x y
z 3 _zscale ActionScript
3D

3D Scaleform Flash 10
+X +Y +Z FOV
55° +Z

3D Flash

- Movie clips
- Text fields
- Buttons

3Di ActionScript

- **_z** - Z 0
- **_zscale** - Z
  100
- **_xrotation** - X
  0
- **_yrotation** - Y
  0
- **_matrix3d** - 16 4×4
  3D NULL 3D
  2D
- **_perspfov** - Field Of View
  0 180 -1
  55°
  FOV

ActionScript gfxExtensions true

_xrotation _yrotation 3D

3Di

例 1

Y 45

```
global.gfxExtensions = true;
mc1._yrotation = 45;
```

例 2

X Z

```
global.gfxExtensions = true;
sq1._zscale = 500;
function rotateAboutXAxis(mc:MovieClip):Void
{
      mc._xrotation = mc._xrotation + 1;
      if (mc._xrotation > 360)
      {
            mc._xrotation = 0;
      }
}

onEnterFrame = function()
{
      rotateAboutXAxis(sq1);
}
```

例 3

3D 3D

```
function PixelsToTwips(iPixels):Number
{
      return iPixels*20;
}
function TranslationMatrix(tX, tY, tZ):Array
{
      var matrixC:Array = [   1, 0, 0, 0,
                        0, 1, 0, 0,
                        0, 0, 1, 0,
                        tX,tY,tZ,1];
```

```
    return matrixC;
}
```

```
this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);
```

4.2 C++から 3Di を利用する

4.2.1Matrix4x4

4×4 row major Matrix4x4

3D
Render_Matrix4x4.h

4.2.2Direct Access

3Di GFx::Value Direct Access
API 3D 3Di
ActionScript

GFx_Player.h [GFx::Value::DisplayInfo]

```
void    SetZ(Double z)
void    SetXRotation(Double degrees)
void    SetYRotation(Double degrees)
void    SetZScale(Double zscale)
void    SetFOV(Double fov)
void    SetProjectionMatrix3D(const Matrix4F *pmat)
void    SetViewMatrix3D(const Matrix3F *pmat)
bool    SetMatrix3D(const Render::Matrix3F& mat)

Double  GetZ() const
Double  GetXRotation() const
Double  GetYRotation() const
Double  GetZScale() const
Double  GetFOV() const
const Matrix4F* GetProjectionMatrix3D() const
const Matrix3F* GetViewMatrix3D() const
bool    GetMatrix3D(Render::Matrix3F* pmat) const
```

4.2.3Render::TreeNode

3D

Render::TreeNode

GFx_Player.h

```
void      SetProjectionMatrix3D(const Matrix4F& m)
void      SetViewMatrix3D(const Matrix3F& m);
```

4.2.4例：Direct Access API を使ったパースペクティブ FOV の変更

```
Ptr<GFx::Movie> pMovie = …;
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
     GFx::Value::DisplayInfo dinfo;
      bOK = tmpVal.GetDisplayInfo(&dinfo);
      if (bOK)
      {   // set perspectiveFOV to 150 degrees
          Double oldPersp = dinfo.GetFOV();
          dinfo.SetFOV(150);
          tmpVal.SetDisplayInfo(dinfo);
      }
   }
```

5 パースペクティブ

Direct Access API
FOV
FOV
0 FOV

z
FOV FOV
0 180 1
179

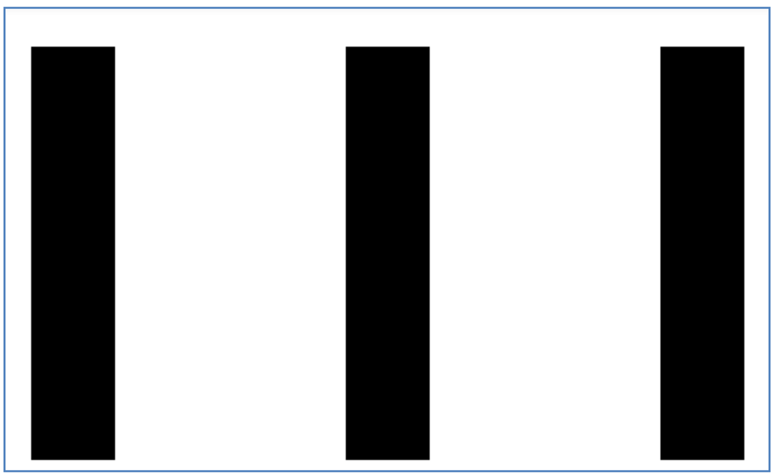

図 2: オリジナルの（回転していない）Flash イメージ

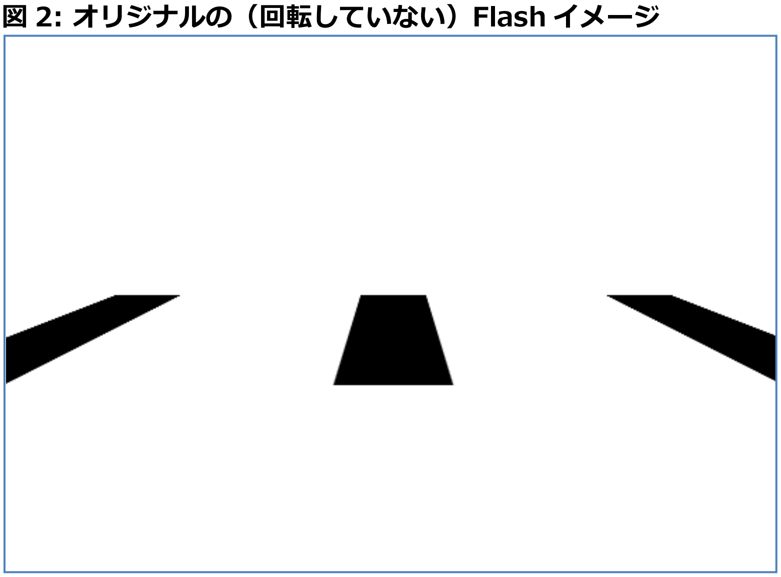

図 3: X 方向に-80 の回転を与え、パースペクティブはデフォルト（55°）

orthographic projection
ActionScript property _perspfov -1
C++ Direct Access API

// 1. Perspective FOV -1

```
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
   GFx::Value::DisplayInfo dinfo;
   bOK = tmpVal.GetDisplayInfo(&dinfo);
   if (bOK)
   {
      dinfo.SetPerspFOV(-1);
      tmpVal.SetDisplayInfo(dinfo);
   }
 }
```

// 2.

```
void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)
{
   const float nearZ = 1;
   const float farZ = 100000;
   float DisplayHeight = fabsf(visFrameRectInTwips.Height());
   float DisplayWidth = fabsf(visFrameRectInTwips.Width());
   matPersp->OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root.mc");
if (bOK)
{
   GFx::Value::DisplayInfo dinfo;
   bOK = tmpVal.GetDisplayInfo(&dinfo);
   if (bOK)
   {
      Matrix4F perspMat;
      MakeOrthogProj(PixelsToTwips (pMovie->GetVisibleFrameRect()), &perspMat);
      dinfo.SetProjectionMatrix3D(&perspMat);
      tmpVal.SetDisplayInfo(dinfo);
  }
}
```

Projection

```
Ptr<Movie> pMovie = …
```

```
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie->GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()),  fabs(visFrameRectInTwips.Height()),
nearZ, farZ);
pMovie->SetProjectionMatrix3D(ortho);
```

5.1 AS3 でのパースペクティブの設定

ActionScript 3　　PerspectiveProjection

Example:

```
import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;
par.y = 100;
par.rotationX = 20;

addChild(par);

var chRed:Sprite = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0, 0, 100, 100);
chRed.x = 50;
chRed.y = 50;
chRed.z = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection = pp3;
```

6 3D での立体画像

Scaleform 3Di 3D UI
2
2

Scaleform 3Di NVIDIA 3D Vision Kit 3D
NVIDIA

Scaleform Display 1 2

P(c)-0 -13(V)-3(ID)-2(IA)]TJC2_1 1 3Tf0 Tc 0 Tw 70929090E±1696B097 -1.272 E±1696906400 61A±16

3D Vision

3D Vision IR CTRL F3
CTRL F4

Scaleform Player
CTRL F5 CTRL F6

NVIDIA *Stereoscopic 3D*
Set up stereoscopic 3D *Set Keyboard Shortcuts* Enable the
advanced in-game settings CTRL F5 CTRL F6
CTRL F7

CTRL F5
10
15
CTRL F5 CTRL F6

CTRL F7

NVIDIA 3D API
3D

API
http://developer.nvidia.com/object/nvapi.html.

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform ステレオ API

Scaleform 4.1 3D Render::HAL
API API 3D
HDMI 1.4

6.2.1 Scaleform

Scaleform 4

Render::Viewport::SetStereoViewport

SF4

```
//

View_Stereo_SplitV        = 0x40,
View_Stereo_SplitH        = 0x80,
View_Stereo_AnySplit    = 0xc0,

void SetStereoViewport(unsigned display);
```

6.2.1Scaleform Stereo 初期化

```
Render::StereoParams s3DInfo;
s3DInfo.DisplayDiagInches = 46;         // 46 inch TV
s3DInfo.DisplayAspectRatio = 16.f / 9.f;
s3DInfo.Distortion = .75;               // 0 to 1
s3DInfo.EyeSeparationCm = 6.4;    // this will default 6.4cm
pRenderHAL->SetStereoParams(s3DInfo);
```

Display 1
2 Display
Scaleform

```
pMovie->Advance(delta);

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);
pMovie->Display();

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);
pMovie->Display();
```

7 サンプルファイル

3Di Scaleform SDK 3D Flash SWF
FLA

C:\Program Files (x86)\Scaleform\GFx SDK 4.2\Bin\Data\AS2or AS3\Samples

図 4: 3DGenerator

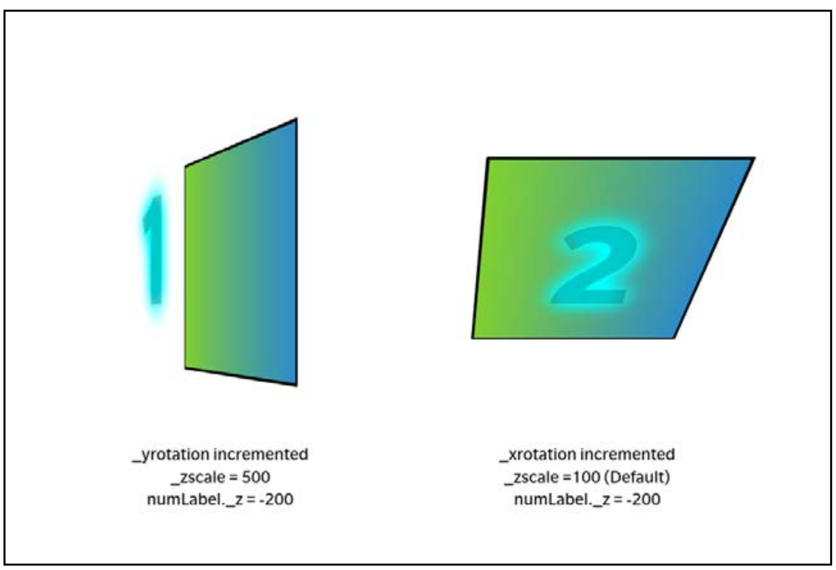

図 5: 3D スクエア


3D Flash - tilted using ActionScript, no render textures used
ScaleformGFx
3D Window
Send
Change Image
Enter Text
Reset
Quit
Scaleform

図 6: 3D ウィンドウ